

The background is a blurred image of a computer monitor. It appears to be a code editor or an IDE, with various windows and panels visible. The text is overlaid on this background.

C#

Level 1

Just In Time Compilation
(JIT)



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Definition

JIT Compiler

-  JIT (Just-In-Time) Compiler is part of the CLR execution environment.

-  It converts IL (Intermediate Language) into native machine code during JIT-based execution.

Explanation

Compilation Stages

-  Source code is first compiled into IL by the C# compiler.
-  When the application runs, JIT converts the required IL into native code.

Execution Flow

-  C# → Compiler → IL → CLR/JIT → Native Code → CPU.
-  The CPU executes the generated native machine code.

Procedures

1. Write C# source code.
2. C# compiler converts it into IL and metadata.
3. The assembly is loaded by the CLR.
4. JIT compiles required IL into native code.
5. CPU executes the native code.

💡 Example

C# Source Code



C# Compiler



IL



JIT



Native Code



CPU

➡ Result

The C# program is converted from IL into machine-specific code that can be executed by the processor.

Note

- ◆ IL may also be called CIL or MSIL.
- ◆ Native code is machine code designed for the target processor.
- ◆ Traditional materials may describe Pre-JIT, Econo JIT, and Normal JIT, but this is mainly a historical/educational classification.

⊘ Warning

- ◆ The *C#* compiler normally does not directly compile *C#* into CPU machine code.
- ◆ JIT and CLR are not the same thing.
- ◆ Modern .NET also supports AOT compilation.



Comparison

▨ JIT

- ✓ Converts IL into native code during execution.
- ✓ Can optimize code for the target environment.
- ✓ Works as part of the .NET runtime execution process.

▤ AOT

- ✓ Compiles code ahead of execution.
- ✓ Reduces the need for runtime compilation.
- ✓ Used by modern .NET in supported scenarios.

✦ Summary

- ✓ JIT = Just-In-Time Compiler.
- ✓ Converts IL into native machine code.
- ✓ Works as part of the CLR execution environment.
- ✓ JIT-based execution happens when the application runs.
- ✓ CPU executes the resulting native code.

Thank you 😊